

Webcam Guard

Technical Guide

Panduan Teknis untuk Administrator dan Developer

Versi 0.7.0 · quizaccess_webcamguard

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1. Persyaratan Sistem

Komponen	Minimum
Moodle	4.1+ (2022112800)
PHP	7.4+
Browser	Chrome 90+, Firefox 90+, Edge 90+, Safari 15+
HTTPS	Wajib - getUserMedia() tidak berjalan di HTTP
Webcam	Diperlukan di sisi mahasiswa

Catatan Browser: MediaPipe Face Detection (primary) membutuhkan WASM + SIMD support. Native FaceDetector API (fallback) tersedia di Chrome/Edge saja. Firefox/Safari: fitur deteksi wajah terbatas.

2. Instalasi

Manual:

```
cp -r webcamguard/ /path/to/moodle/mod/quiz/accessrule/webcamguard/  
php /path/to/moodle/admin/cli/upgrade.php
```

Via Git:

```
cd /path/to/moodle/mod/quiz/accessrule/  
git clone https://github.com/adittanu/moodle-plugins.git temp  
cp -r temp/mod/quiz/accessrule/webcamguard/ ./webcamguard/  
php /path/to/moodle/admin/cli/upgrade.php
```

[Screenshot]

Halaman Manage Access Rules — Webcam Guard muncul di daftar

Simpan sebagai: 01-plugin-installed.png

Gambar 1: Verifikasi instalasi plugin berhasil

3. Konfigurasi Global

Lokasi: Site administration → Module settings → Webcam Guard

[Screenshot]

Halaman Admin Settings — LiveKit config + Data Retention

Simpan sebagai: 02-admin-settings.png

Gambar 2: Halaman konfigurasi global Webcam Guard

LiveKit (Opsional)

Setting	Keterangan	Contoh
LiveKit WebSocket URL	URL server LiveKit	wss://livekit.example.com
LiveKit API Key	API key dari dashboard	APIxxxxxxxxxx
LiveKit API Secret	Secret untuk sign token	xxxxxxxxxx
Token TTL	Durasi token live (detik)	300

Data Retention

Setting	Keterangan	Default
Retention period	Berapa lama data disimpan sebelum dihapus otomatis	30 hari

Pilihan: 7, 14, 30, 60, 90, 180, 365 hari.

4. Konfigurasi Per-Quiz

Lokasi: Edit quiz → Webcam Guard section



Gambar 3: Pengaturan Webcam Guard di quiz settings

Setting	Tipe	Default	Keterangan
Enable Webcam Guard	Checkbox	Off	Aktifkan monitoring
Snapshot on violation	Checkbox	On	Ambil foto saat pelanggaran
Interval snapshots	Select	Off	Foto berkala (60/120/300s)
No-face threshold	Number	10 detik	Batas wajah tidak terdeteksi
Multiple-face threshold	Number	3 detik	Batas banyak wajah
Blur threshold	Number	5 detik	Batas pindah tab
Live monitoring	Checkbox	Off	Streaming webcam live
Identity verification	Checkbox	Off	Verifikasi wajah
Identity threshold	Number	60	Nilai 30-90
Identity mode	Select	Flag	Flag atau Block

5. Arsitektur

Plugin mengikuti arsitektur Moodle quiz access rule:

rule.php - Main access rule class

preflight.js - Verifikasi identitas (face-api.js)

monitor.js - Monitoring runtime (MediaPipe + native fallback)

live_student.js - LiveKit publisher (320x240@10fps)

live_dashboard.js - Dashboard multi-attempt guru (max 20 tiles)

locallib.php - Shared helper functions

[Screenshot]

Diagram Arsitektur — Komponen dan alur data

Simpan sebagai: 04-architecture.png

Gambar 4: Arsitektur plugin Webcam Guard

[Screenshot]

Diagram Alur Monitoring — Dari preflight sampai cleanup

Simpan sebagai: 05-monitoring-flow.png

Gambar 5: Alur monitoring dari awal sampai akhir

6. Database Schema

quizaccess_wg_config

Konfigurasi per-quiz. Kolom: quizid (unique), enabled, snapshotonviolation, intervalseconds, nofacethreshold, multifacethreshold, blurthreshold, identityenabled, identitythreshold, identitymode, liveenabled.

quizaccess_wg_events

Event monitoring. Kolom: id, courseid, cmid, quizid, attemptid, userid, eventtype, durationms, severity, metadata (JSON), hassnapshot, clienttime, timecreated.

quizaccess_wg_reviews

Status review guru. Kolom: id, quizid, attemptid, userid, status (pending/cleared/suspicious), reviewedby, reviewcomment.

quizaccess_wg_live

Request live monitoring. Kolom: id, courseid, cmid, quizid, attemptid, userid, requestedby, roomname, status, expiresat.

7. Web Service API

Function	Method	Capability	Keterangan
quizaccess_webcamguard_log_event	Write	mod/quiz:attempt	Mahasiswa kirim event
quizaccess_webcamguard_request_live	Write	viewreport	Guru mulai/stop live
quizaccess_webcamguard_poll_live	Read	mod/quiz:attempt	Mahasiswa cek live request
quizaccess_webcamguard_poll_live_stats	Read	viewreport	Dashboard refresh stats

8. Capabilities

Capability	Role Default	Keterangan
quizaccess/webcamguard:configure	Editing Teacher, Manager	Konfigurasi quiz settings
quizaccess/webcamguard:viewreport	Teacher, Editing Teacher, Manager	Lihat report dan snapshot
quizaccess/webcamguard:reviewattempt	Teacher, Editing Teacher, Manager	Ubah status review

9. LiveKit Integration

Setup: Buat akun LiveKit Cloud → dapatkan URL/Key/Secret → masukkan di admin settings → aktifkan di quiz settings.

Room: wg-q{quizid}-a{attemptid}-{random_hex}. Token: HS256 JWT. Teacher: subscribe-only. Student: publish-only.

10. Scheduled Tasks

Cleanup Task: Setiap hari jam 03:17. Hapus data lebih lama dari retention period. Batch 1000 records/iterasi.

11. Privacy & GDPR

Implementasi lengkap Moodle Privacy API: metadata, export (per user per context), delete, get_contexts_for_userid, get_users_in_context.

12. Troubleshooting

Webcam tidak muncul

- Pastikan HTTPS
- Cek browser console
- Cek izin kamera
- Cek quiz settings

MediaPipe tidak jalan

- Chrome 90+ / Edge 90+ required
- Firefox/Safari: fallback ke native API

Live monitoring tidak aktif

- Cek LiveKit config
- Cek quiz settings live monitoring
- Cek capability viewreport

Data tidak dihapus otomatis

- Cek Scheduled Tasks enabled
- Cek retention setting
- Jalankan manual cleanup

13. File Structure

rule.php, locallib.php, settings.php, attempt.php, report.php, lib.php, version.php, styles.css —
Komponen utama PHP

amd/src/ — 5 JS modules: preflight, monitor, live_student, live_teacher, live_dashboard

classes/external/ — 4 web service APIs

classes/privacy/ — GDPR provider

classes/output/ — HTML builders

classes/livekit/ — JWT token service

classes/task/ — Cleanup scheduled task

db/ — Schema, capabilities, services, tasks, upgrade

lang/en/, lang/id/ — Language strings

faceapi/, mediapipe/, livekit/ — Bundled libraries

Lampiran: Daftar Gambar

Berikut daftar screenshot yang perlu diambil dan disimpan di folder **pix/guide/**:

No	File	Deskripsi	Lokasi Pengambilan
1	01-plugin-installed.png	Verifikasi plugin di Manage Access Rules	Site admin → Plugins → Quiz → Access Rules
2	02-admin-settings.png	Halaman konfigurasi global	Site admin → Module settings → Webcam Guard
3	03-quiz-settings.png	Quiz settings dengan Webcam Guard plugin	Site admin → Webcam Guard
4	04-architecture.png	Diagram arsitektur komponen	Buat manual (diagram/draw.io)
5	05-monitoring-flow.png	Diagram alur monitoring	Buat manual (diagram/draw.io)